GenAI

ANN → CNN
RNN
LSTM
GRU
→ Transformer

LLM ⇒ 1, 2 → Billions and Trillions

→ Daily projects / Company projects

Models → Deepseek
GPT →
⇒ Download

Upload your Model

Hugging face

{ TF Hub, Kaggle }

→ Github of ML
→ Open source
→ Models, Datasets
→ Pretrained

→ Medical Diagnosis ⇒ x-Rays

LLM → Specialise
⇒ Thousands of x-Rays

⇒ Legal Document analysis
⇒ legal contracts

LLM →

→ Financial → Fraud

{Txn → Fraud, Not fraud}

→ Customer - chat → Educorp

FAQs, Terms of service } → LLM

Gaming domain → Sick

Sick — LLM

Slow

Fine Tuning

→ Pre-trained model → Open AI

Google

Deepseek

→ Train further

→ Specific dataset for particular task

→ Retain general knowledge
Refining for specific domains/aspects

→ Optimize for our use

---

10 Billion Param Model

w, b → Floating Point

→ 32 | 16
⇓
Precision

2 Bytes per Param

Store 10 B params → 10 B $\cdot$ 2 Bytes
= $\boxed{20 GB}$

Train → Gradients → 20GB

Optimization → 100 GB

Adam

10 B param Model

1 epoch → 20 GB params + 20 GB gradients + 100 GB opt.

≈ 140 GB

Multiple epochs

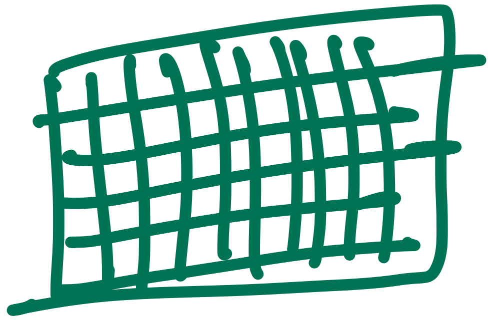
Updating all Params → Training

Full Parameter Fine Tuning X

$Wx + B$

$\downarrow$ wt $\downarrow$

Bias

 $+ \Delta W$

$W \rightarrow (W + \Delta W)$

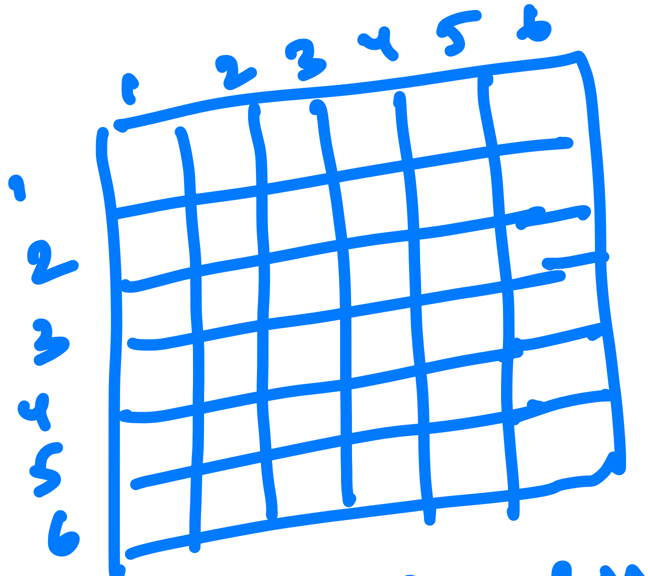
$Wx + B$

$(W + \Delta W)x + B$

PEFT → Parameter Efficient Fine Tuning

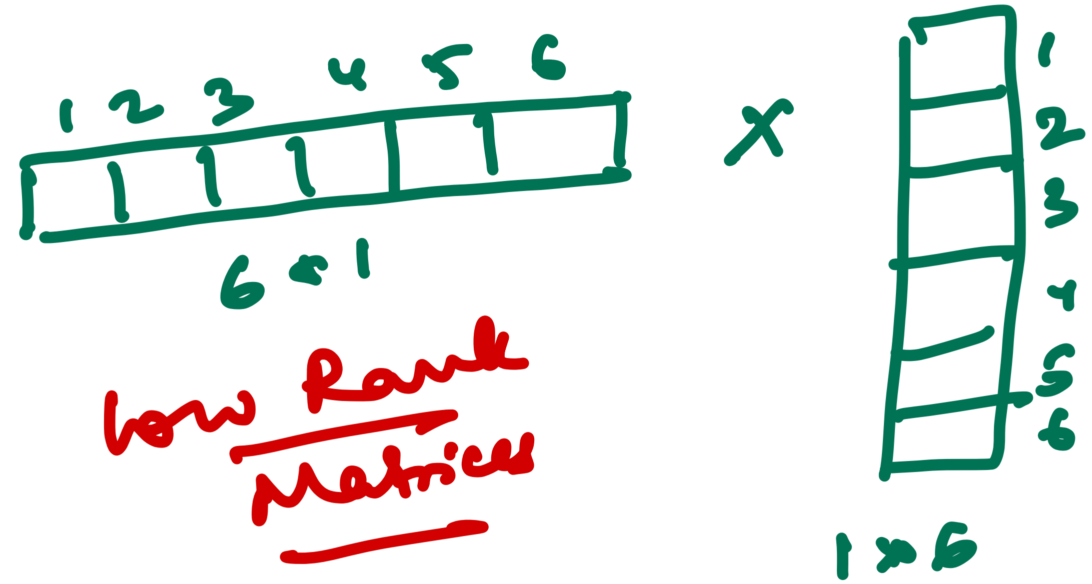
LORA → Low Rank Adaptation

wts matrix → 6 x 6



36 elements

36 parameters



low Rank Matrices

⇓

6 x 6

⇓

36 elements

⇓ ⇑

12 elements

Matrix Decomposition

Q

$2^9 \cdot 2^9 \approx 2^{10}$

$512 + 512$

$= 1024$

$2^{10} \Rightarrow 1024$

$5 * 5$

$5 * 1$

$1 * 5$

No. of params $= 25$

$\Downarrow$

$(5 + 5)$

$\Downarrow$

10 params

25 → 10

Accuracy / Precision

⇓

Computation

LLM

GPT / Claude / Deepseek / Llama

Billion
Trillion

Wts & biases
(params)

Params

update
minimum
no. of
params

Specific
Task

Pre-training

wt changes ⇒ separate smaller matrices]

low rank Matrices

512 * 512 ⇒

Matrix

ΔW

$\Delta W = A \; B$

$A \to 512 * 1$

$B \to 1 * 512$

$W x + B$

$(W + \Delta W) x + B$

$(W + \underbrace{AB}_{\text{low rank Matrices}}) x + B$

$x \to$ i/p

$B \to$ biases

been calculated by pre-training

W → frozen

→ Not making any changes to W

$\boxed{A, B \rightarrow \text{low Rank Matrices}}$

W

A, B

6×6

36 params

Rank=1

Rank=2

Precision

changing Rank

QLoRA

Quantized LoRA

16 bits

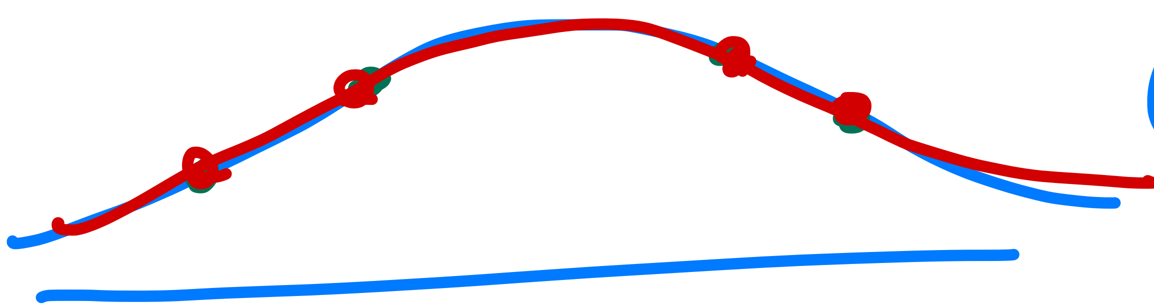
FP 16

4 bits

NF4

bell curve

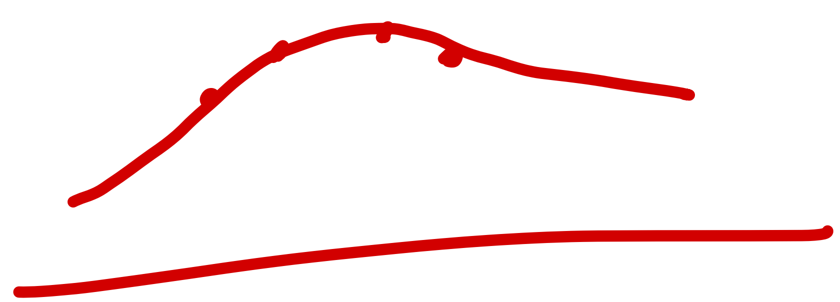
Params



100B

16 bits

4 bits



$(W + \Delta W)x$

$\parallel$

AB

Scaling Factor

$\frac{\alpha}{Rank} = 2$

$W + 2(AB)$

$W + \frac{1}{2}(AB)$

α, Rank

Hyperparameter

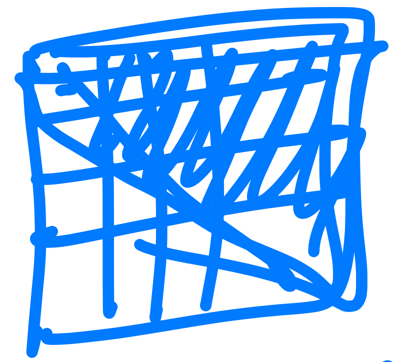
$(W + \frac{AB}{\quad})x + b$

Encoder Contextualized embeddings → BERT

→ no masking

→ future tokens

Decoder

Masked → 

hide future tokens → decoder

encoder

I love learning at Educosys

Bidirectional

BERT → Bidirectional
Encoder
Representation from
Transformers

NLU

Sentiment Analysis

Summarization

QnA

Google Search

Pretraining

MLM → Masked Language Model

Fill in the blanks

Next Sentence Prediction

S — S

Binary classification problem

Self Supervised

Labels